

Maham Rafiq

Computer Scientist — Aspiring Software Engineer

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— LinkedIn — GitLab

Objective

Motivated computer scientist with a strong foundation in programming, object-oriented design, data structures, and database management. Seeking a software engineering internship at **Technosoft**, where I can leverage my skills in programming fundamentals, efficient algorithm design, and database development to contribute to innovative projects and grow professionally.

Education

Bachelor of Science in Computer Science University of Engineering and Technology, Lahore
CGPA: 3.2 — Expected Graduation: 2025

Skills

- **Programming Languages:** Python, C, C++, C, SQL
- **Frameworks:** Angular, MERN, .NET
- **Data Structures Algorithms:** Linked Lists, Trees, Graphs, Sorting Searching Algorithms
- **Database Management:** SQL, MongoDB (Query Optimization, Schema Design)
- **Machine Learning:** Regression, Classification, Clustering, NLP
- **Web Development:** HTML, CSS, JavaScript
- **Version Control:** Git, GitHub, GitLab

Work Experience

Research Assistant *University of Engineering and Technology, Lahore 2023*

- Developed a theft detection system using YOLOv8 for real-time object detection.
- Designed an alarm notification system to alert users during theft detection.
- Focused on computer vision techniques and model optimization for enhanced accuracy.

Web Designer *UET Lahore 2021*

- Designed and implemented user-friendly web interfaces using HTML, CSS, and C.
- Created dashboards that integrated data from multiple sources for academic purposes.

Projects

Home and Habitation

- Scraped real estate data from Zameen.com using web scraping techniques.

- Leveraged data structures and algorithms to process and organize large datasets efficiently.

DBS MidProject

- Designed and implemented a database schema for a mid-term project.
- Focused on optimizing query performance and ensuring data consistency.

DB2021 Final Project

- Created a database system with advanced relational design and indexing techniques.
- Developed stored procedures and triggers for enhanced functionality.

Quiz Application (ASP.NET)

- Developed a quiz application designed for technical learning, including features for data science and computer science topics.
- Integrated SQL database for question management and performance tracking.

Loan Application (Node.js)

- Built a loan application backend using Node.js, implementing RESTful APIs for loan application processing.
- Designed the database schema to handle user accounts, loan statuses, and transactions.

Information Retrieval System

- Developed a document search engine capable of retrieving information based on title, content, and context.
- Applied advanced indexing and search algorithms for efficient retrieval.

Certifications

- **SQL (Advanced)** — 2022
- **Programming in C++** — 2023

Extracurricular Activities

- **Event Host:** Hosted several university events, including the **Closing Ceremony of ITECH 2024**.
- **Graphic Designer:** Worked as part of the graphic design team for the **Software Square Society** of the Computer Science Department.
- **Mentor:** Guided students in mastering programming fundamentals, OOP, and DSA.
- **Member:** Active participant in the Data Science Society, attending workshops on advanced algorithms and database optimization.